Chapter 8

Introduction to Government Revenue

This chapter introduces the fourth, and most extensive, part of this book and shifts the focus from government expenditures to revenues. There are slides and a video associated with this chapter. The following topics are covered:

- Overview of federal, state, and local revenue over time
- Tax terminology
- Tax incidence
- Tax evaluation criteria
- Tax collection steps

Additional information from the Congressional Budget Office (CBO) can be found on their webpage of Major Recurring Reports. Those reports list statistics related to taxation and household income in the United States. In addition, the Internal Revenue Service (IRS) compiles statistics as well at Tax Statistics.

8.1 Introduction

Raising revenue through taxation reallocates scarce resources from the private to the public sector. The revenue enables the funding of services and infrastructure. How the revenue is collected and distributed affects economic outcomes as well as various socioeconomic groups. Tax revenue is redistributed back into the economy, supporting public services, social programs, and investments in national infrastructure. Government revenue does not solely come from taxes but can also include other sources such as payments for services (e.g., purchasing a stamp from the U.S. Postal Service), or fees such as for annual vehicle registration (at the state-level). Income tax is a more traditional form of taxation but is only one of several ways governments can finance their activities. Other activities

besides the one mentioned are privatization of government assets or services and debt financing, each carrying distinct economic and social implications.

At the federal level, revenue primarily comes from personal and corporate income taxes, along with payroll taxes that finance social security. The federal government does not impose sales or property taxes, but a well-known consumption tax are gasoline and diesel taxes.

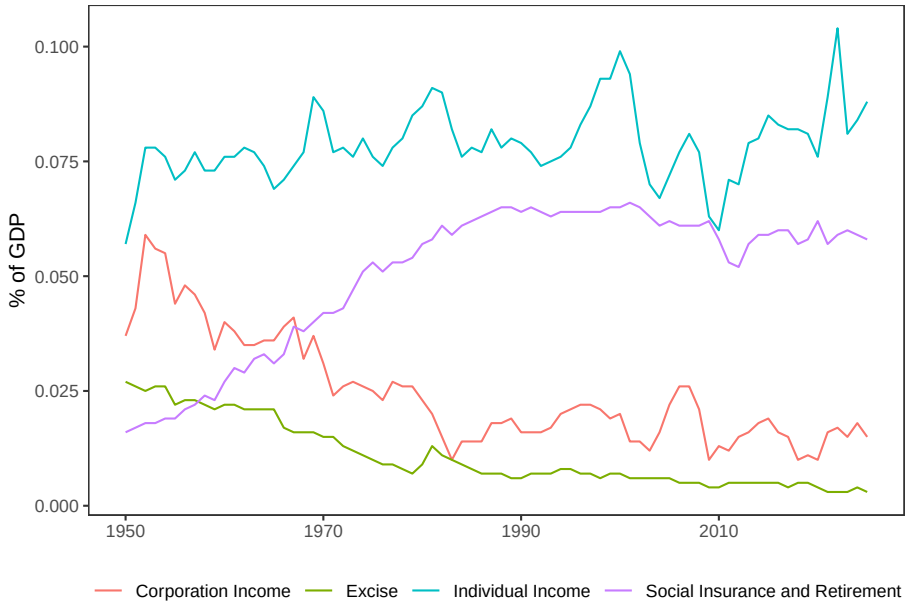


Figure 8.1: Tax Revenue Distribution at the Federal Level

States generally rely on sales and excise taxes as well as personal income taxes for their revenue as illustrated in the graph below. Note that transfers between governments (i.e., federal, state, and local) are intergovernmental transfers and are important sources of revenue for both state and local governments.

Local governments, on the other hand, primarily generate revenue through property taxes, general sales taxes, user fees, and support from state and federal aid (i.e., intergovernmental transfers).

8.2 Tax Terminology

Taxes are compulsory payments associated with certain activities (e.g., earning income, owning property, purchasing goods). They reduce an individuals' ability to allocate resources. Tax policy seeks to achieve that transfer of resources with the least possible economic and/or social harm. Taxes are neither a price for

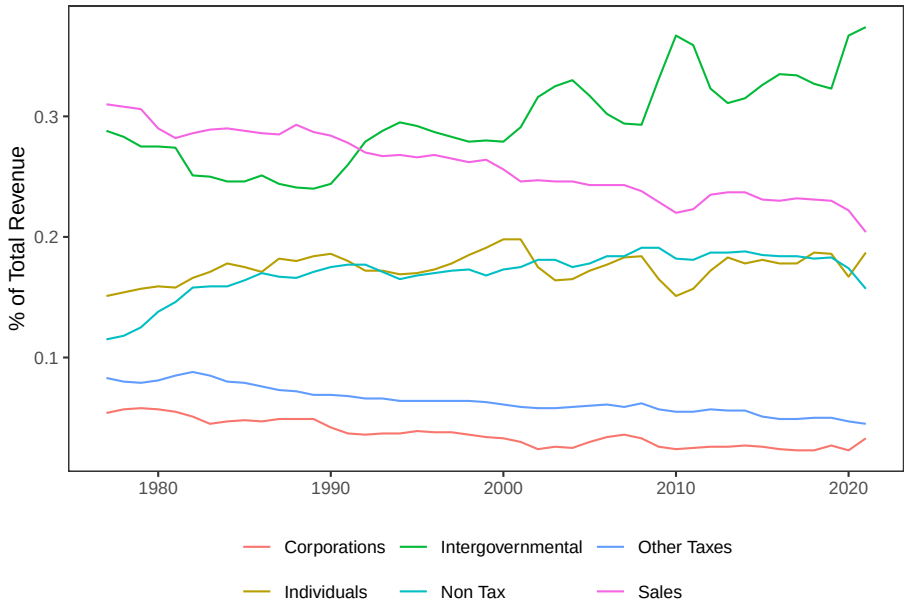


Figure 8.2: Revenue at the State Level

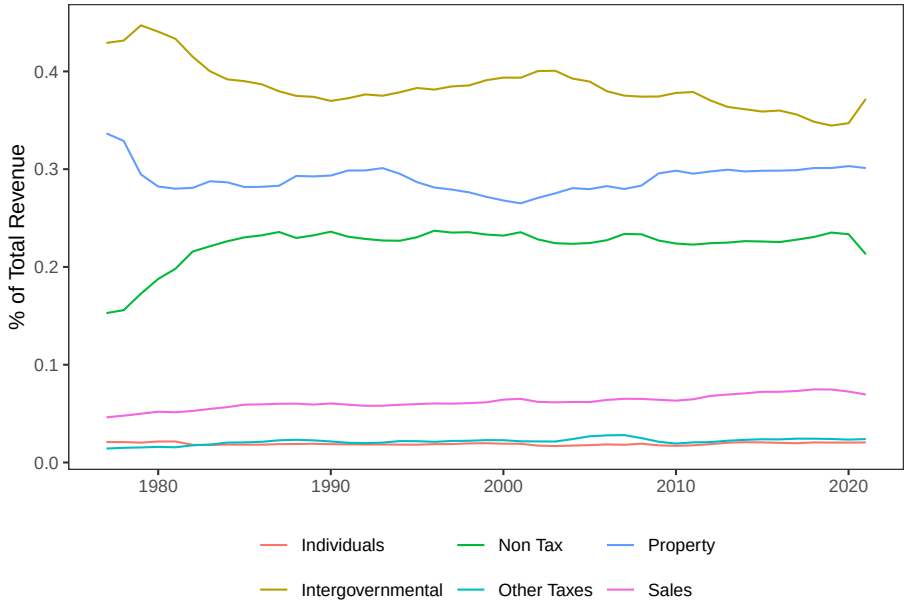


Figure 8.3: Revenue at the Local Level

(direct) services received nor a voluntary contribution.

In his book *The Wealth of Nations*, Adam Smith (1776) establishes the following tax maxims:

- **Equity:** The subject of every state ought to contribute towards the support of the government, as nearly as possible, in proportion to their respective abilities; that is, in proportion to the revenue which they respectively enjoy under the protection of the state.
- **Certainty:** The tax which each individual is bound to pay ought to be certain and not arbitrary. The time of payment, the manner of payment, the quantity to be paid, ought all to be clear and plain to the contributor, and to every other person.
- **Convenience:** Every tax ought to be levied at the time or in the manner, in which it is most likely to be convenient for the contributor to pay it.
- **Economy:** Every tax ought to be so contrived as both to take out and to keep out of the pockets of the people as little as possible, over and above what it brings into the public treasury of the state.

The following sections cover three important tax aspects: (1) Tax base and tax trigger, (2) Tax rate, and (3) Tax incidence.

8.2.1 Tax Base and Tax Trigger

The tax base refers to the item or economic activity on which a tax is imposed. Common examples of tax bases include income, wealth, and consumption. These form the foundation on which various tax systems are built, allowing governments to collect revenue based on these defined economic activities or assets. Taxation can be categorized as either general or selective. A general tax applies to the entire tax base without any exclusions or deductions. In contrast, a selective tax targets only a specific portion of the base, focusing on particular goods, services, or activities. Indiana sales tax is an example of a selective tax. According to the Sales Tax Information Bulletin 29 (August 2023) by the Indiana Department of Revenue, we have the following:

> Generally, the sale of food and food ingredients for human consumption is exempt from Indiana sales tax. Primarily, the exemption is limited to the sale of food and food ingredients commonly referred to as "grocery" food.

The tax trigger is the event or activity that gives rise to the tax liability. This can be determined by the actions of taxpayers, such as earning income or making purchases. Alternatively, the tax trigger can be set by the government, as in the case of property taxes, where the ownership of property creates the tax obligation.

The measurement of the tax base and the determination of tax liability are critical processes in taxation. These involve assessing the size or value of the tax base (e.g., determining the amount of income, wealth, or consumption subject

to tax) and calculating the corresponding tax liability based on the established tax rates and rules.

8.2.2 Tax Rates

Tax rate refers to the rate at which the tax base is taxed. In general, the tax rate applied to the tax base determines how much tax is owed. Let us look at income taxes as an example.

The tax base is income, which is adjusted for exclusions, deductions, exemptions, and credits. The next chapter covers the exact mechanics of those items The tax trigger is engaging in an activity that generates income. The tax rate typically follows a graduated schedule, where different portions of income are taxed at varying rates. That is, a graduated schedule involves marginal tax rates that apply to the last dollar of income earned. There is a distinction between the average, marginal, and effective tax rate.

The average tax rate is the total tax collected divided by the dollar value of the tax base, giving an overall sense of the taxpayer's burden. The marginal tax rate is the change in taxes collected divided by the change in the value of the tax base, showing how much additional tax is paid on the next unit of income. The effective tax rate is the tax collected divided by a measure of ability to pay, such as total income or wealth. This rate is useful in evaluating how taxes impact individuals or entities relative to their overall financial position.

The effective tax rate is often used as an indicator to differentiate between progressive, proportional, and regressive tax systems. For example, in a progressive tax system, the effective tax rate increases with income. In contrast, the effective tax rate remains constant in a proportional tax system, and it decreases with income in a regressive tax system. In the income tax table below, the average and effective tax rates are identical.

	A	B	C
Income	\$10,000	\$40,000	\$100,000
Proportional tax			
Tax amount	\$1,000	\$4,000	\$10,000
Effective tax	10.0%	10.0%	10.0%
Progressive tax			
Tax amount	\$1,000	\$5,000	\$20,000
Effective tax	10.0%	12.5%	20.0%
Regressive tax			
Tax amount	\$1,000	\$3,000	\$6,000
Effective tax	10.0%	7.5%	6.0%

Next, consider the following marginal tax rate table. We are going to use this table for the tax calculations that follow.

Over	But not above	Over	But not above	Tax Rate
Single		**Married**		
$-	$9,525	$-	$19,050	10%
$9,526	$38,700	$19,051	$77,400	12%
$38,701	$82,500	$77,401	$165,000	22%
$82,501	$157,500	$165,001	$315,000	24%
$157,501	$200,000	$315,001	$400,000	32%
$200,001	$500,000	$400,001	$600,000	35%
$500,001		$600,001		37%

Let us consider the following situation of a married couple with household income of $137,000. Note that in the U.S., households pay income taxes and not individuals. Based on the above marginal tax rate table, the tax liability is calculated as follows:

$$19050 \cdot 0.1 + (77400 - 19050) \cdot 0.12 + (137000 - 77400) \cdot 0.22 = 22019$$

The tax liability amounts to $22,019 and the marginal tax rate it 22%. The average tax rate is 16% (i.e., $22,019 divided $137,000).

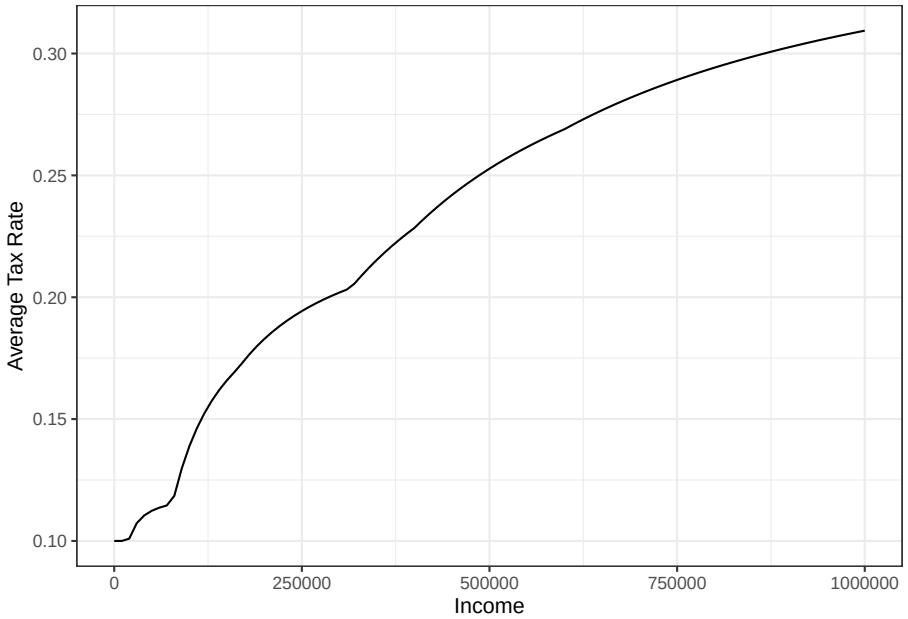


Figure 8.4: Marginal versus Effective Rate

8.3 Tax Incidence

Tax incidence refers to the person or entity that ultimately pays the tax. This can differ from who is legally responsible for submitting the tax to the government. For example, a property tax on rental properties may be legally paid by the landlord but the economic burden falls on tenants through higher rent. Similarly, payroll taxes might be paid by the employer, but the burden could shift to employees through lower wages. The legal or statutory incidence refers to the party that is legally obligated to remit the tax. For example, employers are responsible for remitting payroll taxes, while property owners are required to pay property taxes. Economic incidence is the actual distribution of the tax burden, which may not align with the legal incidence. It refers to who ultimately bears the economic burden of the tax after market adjustments, such as price changes. For a consumption tax, the distribution of the tax burden depends on the relative elasticity of supply and demand. If the demand is more inelastic than supply, consumers will bear more of the tax burden. Conversely, if supply is more inelastic than demand, producers will bear a greater share of the burden.

8.4 Tax Evaluation Criteria

Taxes have a significant effect on the private economy and households. Key considerations include equity and efficiency of different taxes. For example, policymakers must assess whether certain taxes place a disproportionate high burden on low-income taxpayers. Additionally, they must consider the impact that taxes have on consumer and business decisions. Tax evaluation involves five criteria to assess the effectiveness and fairness of the tax system: (1) Economic efficiency, (2) Equity, (3) Adequacy, (4) Feasibility, and (5) Transparency. Feasibility can be further divided into administrative and political feasibility.

8.4.1 Economic Efficiency

Economic efficiency in taxation is related to market equilibria, which represent points where consumer and producer surplus (with social welfare as the sum of the two) is maximized. In an efficient market, goods are produced according to consumer preferences, at the lowest cost, and distributed based on willingness-to-pay. However, taxes can cause deviations from this equilibrium, potentially reducing efficiency. Instances of market failure represent exceptions to this principle, where taxes might address restore efficiency such as the case of a corrective tax on externalities.

Let us consider the example of a sales tax on bread. From a consumer perspective, the introduction of a sales tax on bread leads to substitution away from bread towards alternatives such as rice and potatoes. Those are less desirable by consumers because otherwise, they would have purchased them before in higher quantities. From a producer perspective, the taxes changes the after-tax price (price received by the producer versus price paid by the consumer) of bread.

In addition to the reduction in demand, producers will be worse off as well. If producers and consumers did not react to a tax, there would not be an efficiency loss.

The questions, which need to be consider are usually the following:

- What economic decisions will this tax change? All modern taxes cause some change in economic behavior.
- How much will consumers and producers change their behavior in response to this tax?

The answer to the second question depends on whether there are good alternatives or substitutes to the good which is taxed. The more substitutes there are, the more the efficiency loss from a tax.

8.4.2 Equity

Usually, there are multiple objectives from imposing taxes. Equity in taxation focuses on fairness issues while achieving those objectives. Tax structures often incorporate equity considerations, though these can involve trade-offs (e.g., among various socioeconomic groups) or lead to unintended consequences. Two main standards of equity are the ability-to-pay principle and the benefits principle.

The ability-to-pay principle states that tax payments should be based on a person's financial capacity, often measured by income or wealth. Within this framework, there are two notions.

Horizontal equity for which individuals with the same ability-to-pay should pay the same effective tax rate. Income as an easy measure for ability to pay but other factors that might influence "ability-to-pay" include wealth, number of children, medical expenses (from varying health conditions), and lifetime earning potential, among others.

And vertical equity for which individuals with different abilities to pay should face different effective tax rates. It addresses how the effective tax rate evolves with income. As mentioned previously, a tax system can be progressive (i.e., rate increasing with income), regressive (i.e., rate decreasing with income), or proportional (i.e., rate constant with income).

An equitable tax system is one in which each taxpayer contributes in line with the amount of benefits they receive from the service provided. The benefits principle posits that the cost of a government-provided good or service should be paid by the taxpayers who benefit from it. Payments are proportional to the benefits received. For example, a family with two children should pay a higher property tax (which is in part used for education) than a family with one child. Those could also be (more or less) proportional to the damage caused. Consider fuel taxes for example that are dedicated to road maintenance. The higher the weight on an axle, the more damage is caused to the road. Thus, fuel taxes based on the amount of fuel consumed, affects people with lower fuel efficiency

more. That lower fuel efficiency is usually caused by heavier vehicles that results in more road damage. Arguments in favor of this principle suggest it aligns with intuitive fairness and encourages conservation of public services. However, it faces challenges in application, as few taxes strictly adhere to this principle, especially for public goods. Moreover, individuals less able to pay may still deserve access to public services. In addition, general taxes such as on income cannot be linked to specific services. Taxes based on the benefits principle do not redistribute income and would probably encourage governments to collect revenue in the form of user fees for many basic services. Benefit principle has been used as a justification for local user fees for services such as water, sewer, garbage, etc.

8.4.3 Adequacy

Adequacy in a tax system refers to its ability to generate sufficient revenue at reasonable rates. Broad-based taxes, like those on income or sales, are preferable as they can raise substantial revenue with lower rates. Revenue fluctuations, such as those tied to gasoline prices, affect the adequacy of tax systems. Reasonable tax rates are crucial for two reasons:

1. It's unfair and unpopular to tax one sector disproportionately, as seen in the hypothetical case of New York raising most of its revenue from a bread tax, which would burden consumers and producers.
2. Increasing tax rates too much can reduce total revenue as people find ways to avoid the tax.

Governments prefer stable revenue sources, meaning minimal fluctuations year over year. For example, a 20% annual swing in property tax revenue would complicate budgeting and might force borrowing or service cuts. However, there's a trade-off between stability and elasticity. An elastic tax, which responds to changes in its base, can result in unstable revenue when the base shrinks. Ideally, taxes should grow with a stable tax base.

8.4.4 Feasibility

Administrative feasibility relates to the ease of tax collection. This includes administrative costs, or the effort required by the government to collect taxes, and compliance costs, which represent the effort taxpayers must exert to meet their tax obligations. Political feasibility focuses on the public's tolerance of taxes. Factors like tax intolerance (the rejection of new taxes), sticker shock (negative reactions to significant tax increases), and reasonableness (the perceived fairness of taxes relative to the value received) play crucial roles. Additionally, visibility affects tax popularity; lower visibility taxes are less likely to face opposition. Exportation, or the perception that outsiders contribute to the tax burden, can increase the acceptance of taxes within a jurisdiction.

8.5 Tax Collection Steps

The collection of taxes—or more accurately, the entire tax system—involves several steps:

1. Compilation and maintenance of taxpayers information
2. Declaration or assessment (tabulation of taxable activity)
3. Revenue and taxpayer accounting
4. Delinquency control, compliance, and collections
5. Audits

The compilation and maintenance of taxpayer information involve tracking identifiers such as social security numbers, taxpayer identification numbers, and parcel numbers for properties such as homes. This information is important to manage tax-related activities and ensuring accurate records for each taxpayer. The declaration or assessment process refers to the tabulation of taxable activities. For the income tax case, this includes sources like miscellaneous income, interest income, dividends, and distributions besides the main source of income.

Revenue and taxpayer accounting is a critical step where payments must be deposited, recorded, and properly distributed. After taxpayers submit their returns, the relevant information must be posted and made accessible to taxpayers for review. Delinquency control, compliance, and collections ensure that taxpayers meet their obligations. Active compliance requires taxpayers to understand their responsibilities, with accessible resources available to assist them in meeting legal requirements. Audits are conducted to ensure that taxpayers comply with tax laws. This process helps detect and address discrepancies, ensuring proper taxation and collection.

8.6 Exercises

1. ***State Tax Revenue*** (***): Download the following files: State Tax Collection (Comma Separated Value File) and State Tax Collection (U.S. Census PDF). The first file contains the tax revenue by state and year for various taxes in real and nominal terms. The PDF-file contains explanatory notes on the Excel file (although the content of the Excel file is relatively self-explanatory). Select a state of your choice (not Indiana) and answer the following questions:

    a. What are the three major tax revenue sources used by your state government in the most recent year? Hint: For each tax, calculate tax revenue as a share of total tax revenue. Select the three taxes with the largest shares.
    b. Use the three major tax revenue sources you identified in part (a) to answer the following questions: How rapidly have those taxes grown between 1994 and the most recent year? What may explain the trends you observe? Hint: Calculate the simple year-to-year growth rate for

each. Plot a graph of the growth rates against time between 1994 and the most recent year.